

Terraform Variables vs Terraform.tfvars

What is `variables.tf`?

The `variables.tf` file is used to **declare variables** in Terraform. It defines the variable name, type, description, default value, and validation rules.

Example:

```
variable "aws_region" {  
  description = "AWS Region"  
  type        = string  
}  
  
variable "instance_type" {  
  description = "EC2 Instance Type"  
  type        = string  
}
```

In this file, we are only defining the variables. No actual values are assigned.

What is `terraform.tfvars`?

The `terraform.tfvars` file is used to **provide values** to the variables declared in `variables.tf`.

Example:

```
aws_region    = "ap-south-1"  
instance_type = "t2.micro"
```

Here, the actual values are assigned to the variables.

Why Do We Use Both?

Terraform separates the **configuration logic** from the **configuration values**.

variables.tf

Defines:

- Variable names
- Data types
- Descriptions
- Validation rules

terraform.tfvars

Stores:

- Actual values
- Environment-specific configurations
- Custom settings

This separation makes Terraform code reusable and easier to maintain.

Can We Use Only variables.tf?

Yes.

Terraform allows default values inside the variable declaration.

Example:

```
variable "aws_region" {  
  type    = string  
  default = "ap-south-1"  
}
```

In this case, Terraform will use the default value if no value is provided elsewhere.

Then Why Use terraform.tfvars?

For small learning projects, using only default values is completely fine.

However, in real-world projects, different environments require different values.

Example:

Development

```
instance_type = "t2.micro"
```

Staging

```
instance_type = "t3.small"
```

Production

```
instance_type = "t3.medium"
```

Instead of modifying the code every time, we simply provide different `.tfvars` files.

This keeps the code clean and reusable.

Real-World Example

variables.tf

```
variable "instance_type" {  
  type = string  
}
```

dev.tfvars

```
instance_type = "t2.micro"
```

prod.tfvars

```
instance_type = "t3.medium"
```

main.tf

```
resource "aws_instance" "web" {  
  instance_type = var.instance_type  
}
```

Now the same Terraform code can be used for multiple environments.

Common Commands

Validate Configuration

```
terraform validate
```

Format Terraform Files

```
terraform fmt
```

Initialize Terraform

```
terraform init
```

Create Execution Plan

```
terraform plan
```

Apply Changes

```
terraform apply
```

Destroy Infrastructure

```
terraform destroy
```

Apply Using a Specific Variable File

```
terraform apply -var-file="dev.tfvars"
```

```
terraform apply -var-file="prod.tfvars"
```

Interview Answer

variables.tf is used to declare variables, including their type, description, and validation rules.

terraform.tfvars is used to assign actual values to those variables.

In simple terms:

- `variables.tf` = Variable Definition
- `terraform.tfvars` = Variable Values

This separation improves reusability, maintainability, and environment-specific configuration management in Terraform.